Unit NFC

SKU:U216



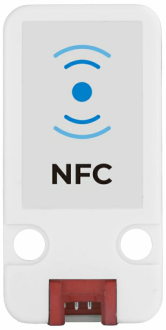
























Description

Unit NFC is a near-field communication read/write unit operating at 13.56 MHz. The module features the ST25R3916 high-performance NFC front-end IC, supporting mainstream protocols including ISO 14443A, ISO 14443B, FeliCa™, and ISO 15693, enabling reliable read/write operations and data exchange with a wide variety of NFC/RFID tags and cards. The IC supports reader/writer mode, card emulation mode, and custom protocol mode, with automatic antenna tuning, high-sensitivity reception, and comprehensive protocol handling capabilities.

This unit communicates with the host device via I2C interface, ensuring reliable and stable data transmission. The product features a LEGO-compatible hole design for easy structural integration and screw-mount installation, making it suitable for a wide range of near-field communication and identity verification applications, including access control systems, identity authentication, and smart transportation.

Features

- 13.56 MHz high-frequency near-field communication
- Equipped with ST25R3916 high-performance IC
- Supports ISO14443A, ISO14443B, FeliCa™, and ISO15693 protocols
- Supports three operating modes: Reader/Writer / Card Emulation / Custom Protocol
- I2C communication interface
- LEGO-compatible mounting holes
- Development Platform
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Unit NFC

- 1 x HY2.0-4P Grove Cable (20cm)

Applications

- Access Control Systems
- Identity Verification
- Smart Transportation

Specifications

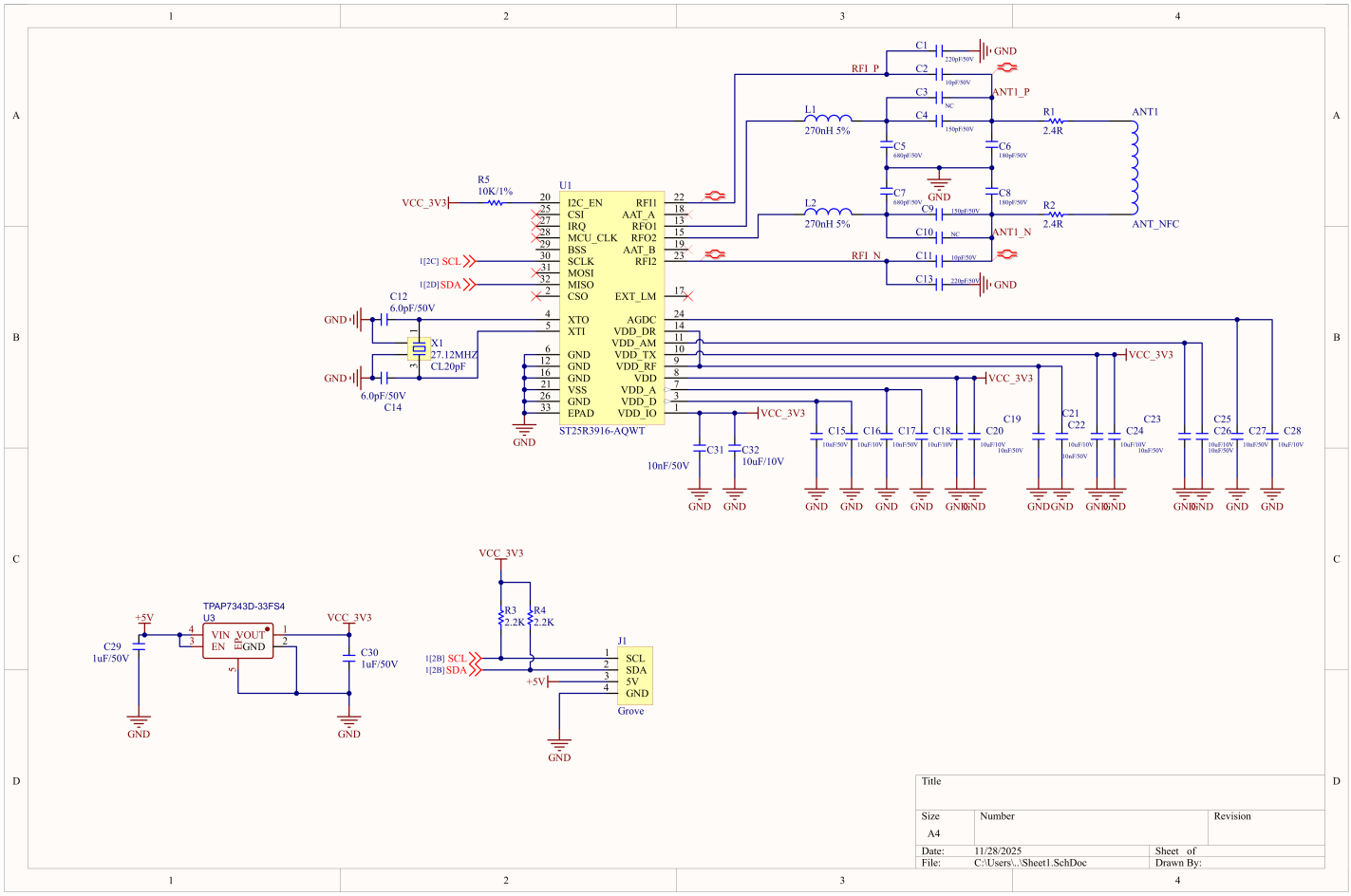
Specification	Parameter
IC Type	ST25R3916-AQWT
Operating Frequency	13.56 MHz
Communication Interface	I2C @0x50 (100K / 400K)
Supported Protocols	**Reader/Writer Mode**: NFC-A/B (ISO 14443A/B compatible), NFC-F (FeliCa™), NFC-V (ISO 15693) **Card Emulation Mode**: NFC-A / NFC-F
Read/Write Range	≤ 40mm
Power Consumption	Sleep: 5V @ 19.33uA Power-on (unconfigured): 5V @ 19.79uA Continuous reading: 5V @ 67.65mA Intermittent reading: 5V @ 66.48mA
Product Size	48.0 x 24.0 x 8.0mm
Product Weight	5.9g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	11.5g

Certifications

- CE/FCC/RoHS/WEEE/MIC

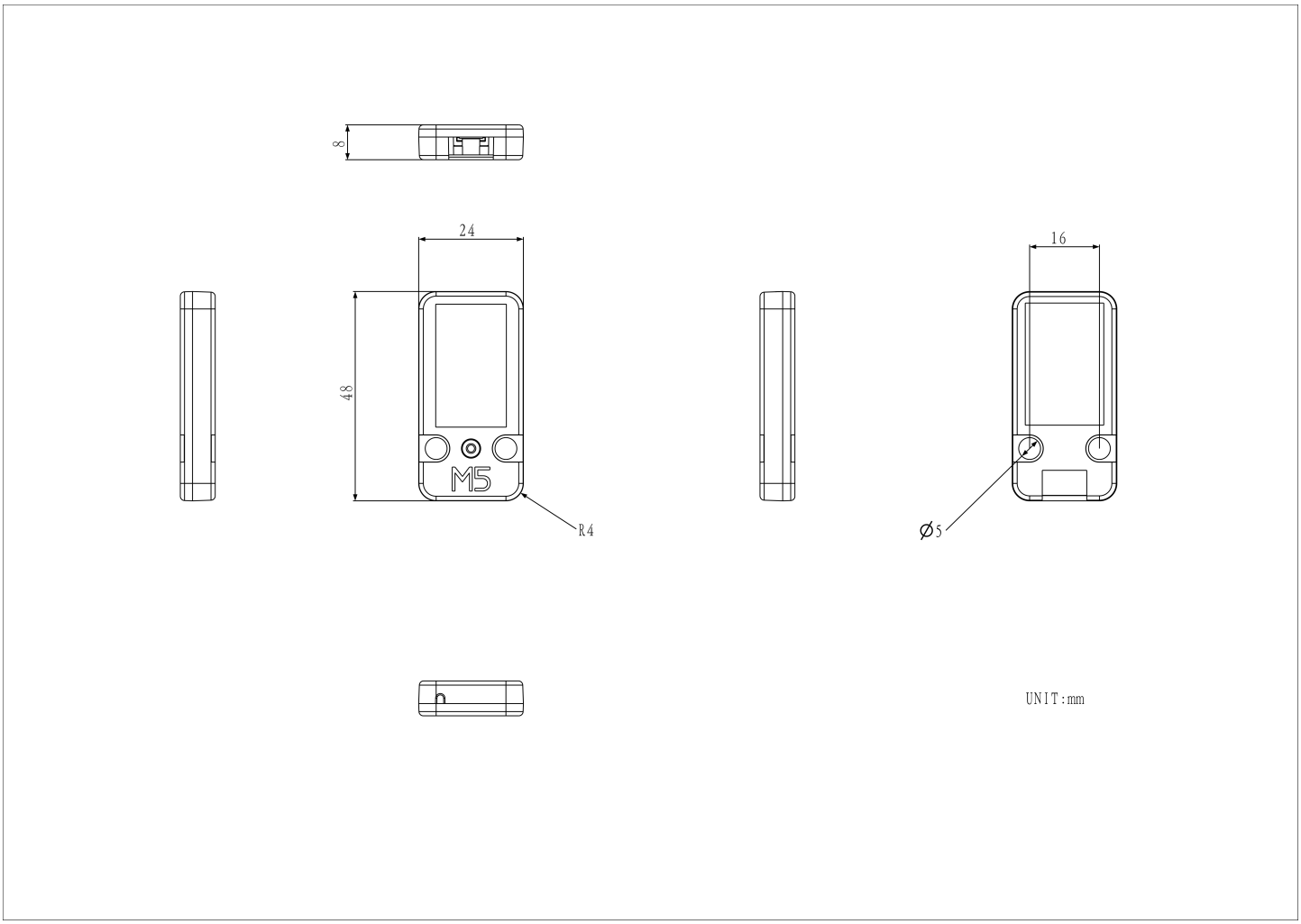
Schematics

- Unit NFC Schematics PDF



Model Size

- Unit NFC Model Size PDF



Structure

- Unit NFC Structure Files

PinMap

Unit NFC

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Softwares

Arduino

- Unit NFC Arduino Library
- Unit NFC Arduino Quick Start

Datasheets

- ST25R3916

Video

- Unit NFC Product Introduction and Functional Demonstration

U216-Unit-NFC-video-EN.mp4